



Spatial patterns of sediment phosphorus contents and release in a deep clearwater lake undergoing rapid eutrophication

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Abstract Symptoms of eutrophication are increasingly evident in remote clearwater lakes. To identify the sources and dynamics of phosphorus release, we measured total phosphorus (TP) in sediments and soluble reactive phosphorus (SRP) fluxes across the sediment–water interface at 54 locations in a deep temperate lake. Once renowned for its clear waters, Lake Stechlin has experienced a fourfold increase in water column TP over the past decade. SRP fluxes from sediments generally increased with water depth across all three lake basins, although there were

significant variations in SRP concentrations, up to threefold, among sampling locations at the same depth. Notably, the lake’s total mean SRP flux in June ($1.12 \text{ mg m}^{-2} \text{ day}^{-1}$) was higher than that determined in October ($0.74 \text{ mg m}^{-2} \text{ day}^{-1}$). This result can be attributed to the substantial contribution (about 31% of total SRP release) of shallower sediments (0–20 m), which are not affected by seasonal anoxia. Our findings highlight a notable spatial variability of SRP fluxes and underscore the importance of considering often overlooked shallow sediments when assessing P dynamics in lakes.

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Introduction

Numerous lakes worldwide have long experienced anthropogenic nutrient inputs related to agricultural practices or industrial and municipal wastewater discharge (Schindler et al. 2016). The resulting eutrophication has been a prime focus of limnological research for more than a century, and the underlying mechanisms are well understood. Phosphorus (P), in particular, plays a critical role as a limiting nutrient regulating primary productivity of most lakes (Gonsiorczyk et al. 2024; Schindler et al. 2016; Stoddard et al. 2016; Vadeboncoeur et al. 2021).

However, eutrophication symptoms have been increasingly observed also in remote lakes, where external nutrient loading is small (Gonsiorczyk et al. 2024; Schindler et al. 2016; Stoddard et al. 2016; Vadeboncoeur et al. 2021). The mechanisms driving this trend are uncertain, although large-scale phenomena including global climate change and long-distance atmospheric transport could play a role (Musolff et al. 2017; Radbourn et al. 2020).

Eutrophication can potentially also be driven by internal P loading, as substantial amounts of P are stored in lake sediments (Steinman and Spears 2020). Redox conditions play a key role in the release of P from these sediments. One well-documented mechanism involves the dissolution of iron oxyhydroxide (FeOOH) complexes under anoxic conditions, which leads to the release of previously sorbed P (Einsele 1936). Consequently, fluctuating redox conditions are often seen as the primary factor controlling P retention versus release at sediment surfaces. However, this process is not universally applicable, as it has become evident that sediment P dynamics are also influenced by a variety of geochemical, physical, and microbiological factors (Boström et al. 1988; Hupfer and Lewandowski 2008). In addition to the release of Fe-bound P, significant amounts of organic P can be mobilized during both aerobic and anaerobic decomposition processes, which begin during the settling of organic material, such as algal detritus, through the water column (Albright and Wilkinson 2022; Kleeberg 2002; Pettersson 2001). The mineralization of sinking organic matter is particularly enhanced under aerobic conditions (Moore et al. 1992), highlighting the contrasting roles of oxygen availability in redox-sensitive P release from sediments and the remineralization-driven release of organic P. Increasing evidence points to internal P loading occurring even under oxygenated water conditions (e.g., Albright and Wilkinson 2022), challenging traditional assumptions about P release dynamics and emphasizing the importance of assessing the spatial and seasonal variations in sediment P pools and fluxes (Spears et al. 2007).

Despite exceptions to the rule (e.g., Mothes and Proft 1985; Tammeorg et al. 2024), calculations of lake-wide P release dynamics are typically based on data from a single or a few sampling locations (Gonsiorczyk et al. 2003; Scholtysik et al. 2020; Yin

et al. 2023) assumed to be representative for the lake as a whole. This approach ignores important variation in sediment characteristics such as texture, redox state or organic matter content and quality, some of which may vary randomly, whereas others change systematically along depth gradients (Brannon et al. 2023; Hupfer and Lewandowski 2008; Tammeorg et al. 2024). Similarly, oxygen depletion is greatest in deep water, leading to depth-dependent changes in oxygen concentrations at the sediment surface and resulting redox gradients. In addition, these redox gradients may vary seasonally. Robust assessments of P release rates from lake sediments are thus likely to benefit from repeated samplings at different water depths.

The objective of the present study was to determine the spatio-temporal variation in the P content of, and P fluxes from sediments in a deep, formerly nutrient-poor clearwater lake. Specifically, the aim was to illuminate the role of sediments of shallow vs. deep water in releasing P to the lake water column and, by extension, in the rapid eutrophication of the lake. We expected to find systematic patterns of P dynamics reflecting basin morphology, particularly due to the dependence of oxygen conditions on water depth. Therefore, we placed particular emphasis on (i) the influence of anoxic conditions on P contents and fluxes in profundal sediments, and (ii) the extent of variation with water depth in different lake basins.

Materials and methods

Study site

The study was conducted in Lake Stechlin, a deep hardwater lake in northeastern Germany, approximately 80 km north of Berlin. The lake's 12.6 km² watershed, shaped by the last glacial period, is primarily covered by managed Scots pine forests and has no major surface in- or outflows, relying on ground-water and precipitation (Casper 1985). Lake Stechlin covers 4.23 km² and has an average water depth of 23.3 m, with its three subbasins ranging in depth from 35 to 69.5 m (Kröger et al. 2023; Otto et al. 2023). Renowned for its high water transparency, with occasional Secchi depths exceeding 10 m, the lake experienced a rise in total phosphorus (TP) concentration from approximately 15 µg L⁻¹ (1990–2010) to over

60 $\mu\text{g L}^{-1}$ by 2020, resulting in increased phytoplankton biomass and changes in community composition (Gonsiorczyk et al. 2024; Kröger et al. 2023; Otto et al. 2023). Oxygen declines in the hypolimnion caused seasonal and interannual expansions of anoxic zones (Gonsiorczyk et al. 2024; Scholtysik et al. 2020).

Sampling and sample analyses

A total of 54 sites were cored in early summer (June) and again in late fall (October) 2021, towards the end of summer stratification. Two randomly positioned transects, perpendicular to the shoreline, were designated in each of the three lake basins to collect sediment cores at 10-m depth intervals starting and ending at 5 m and crossing the deepest point in each basin (Fig. 1a). This resulted in four depth equivalent cores at each depth in each basin, except for the deepest point where single samples were taken. A boat equipped with a GPS controlled i-Pilot for an electrical trolling motor (Minn Kota Powerdrive 55, Racine, WI, USA) was used to ensure the boat precisely held the desired position during coring. The GPS coordinates of the sampling points in June were recorded to return to the same sampling points, within a radius of 5 m, in October.

Sediment cores were collected using clear PVC tubes (60 cm long, 6 cm diameter) attached to a Kajak gravity corer (UWITEC GmbH, Mondsee, Austria). Clear water (~ 2 mL) was sampled 1 cm above the sediment surface with a syringe and silicone tube, filtered through a 0.45 μm membrane, and stored at 4 °C until analysis the next day. Porewater was extracted at the lakeside laboratory using 5 cm CSS Rhizon samplers (0.15 μm pore size, Rhizosphere Research Product B.V., Wageningen, The Netherlands), inserted into pre-drilled holes in the PVC tubes to collect 2 mL of filtered porewater from ~ 2 cm below the sediment surface.

Sediment wet and dry mass were determined by weighing the surface sediment samples to the nearest 1 mg before and after freeze-drying. Subsequently, the sediment samples were ground in a ball mill before determining organic carbon content (C_{org}) with a Leco RC612 multi-phase carbon analyzer (Leco Corp., St. Joseph, MI, USA). Contents of TP in the surface

sediments (0–2 cm) were determined after digesting 4.0 to 5.0 mg of ground subsamples with 0.18 mM $\text{K}_2\text{S}_2\text{O}_8$ for 30 min at 134 °C in an autoclave (Krom and Berner 1980). The sedimentary TP pool was analyzed only once in June, assuming that, in contrast to the soluble P fraction, no significant changes would occur within a few months. P concentrations were determined spectrophotometrically using a flow injection analyzer (FIA Star 5010, Foss Tecator, Höganäs, Sweden). SRP in filtered (0.15 μm pore size) porewater and water overlying the sediment sampled in June and October was analyzed on the same instrument without prior sample digestion.

Vertical profiles of temperature and O_2 concentrations were measured with a YSI multiprobe (YSI 6600, Yellow Springs, OH, USA). Water samples for the SRP and TP analyses were taken with a Limnos sampler at a vertical distance of 5 m. SRP was determined in filtered water samples (cellulose acetate, 0.45 μm , Sartorius, Göttingen) and TP sample after digestion with $\text{K}_2\text{S}_2\text{O}_8$ (134 °C, 30 min) as described above.

Data analysis

SRP fluxes were derived from concentration gradients across the sediment–water interface (SWI) according to Fick's first law of diffusion (Boström et al. 1988; Nowlin et al. 2005; Syers et al. 1973):

$$J_{SRP} = \frac{\varphi}{\theta^2} \times D_{SRP} \times \frac{\Delta C_{SRP}}{\Delta z},$$

where J_{SRP} is the flux of SRP across the sediment surface ($\text{mg m}^{-2} \text{d}^{-1}$), φ is the sediment porosity, θ is the sediment tortuosity, D_{SRP} is the molar diffusion coefficient, $\Delta C_{SRP}/\Delta z$ is the concentration gradient at the SWI between the overlying water and the mean sediment sampling depth of 1 cm (Lewandowski et al. 2002).

The dry sediment density (ρ_{dry}) was calculated as:

$$\rho_{dry} = \frac{1}{\frac{g}{\rho_{org}} + \frac{1-g}{\rho_{min}}},$$

where g is the organic matter content (wt. %) calculated as $2 \times C_{org}$, ρ_{org} is the organic matter density (1.40 g cm^{-3}) and ρ_{min} is the density of mineral matter (2.65 g cm^{-3}). These density values were adopted

from Redding et al. (2005). The density of wet sediment (ρ_{wet}) was calculated as:

$$\rho_{wet} = \frac{1}{\frac{w}{\rho_{water}} + \frac{1-w}{\rho_{dry}}},$$

where w is the water content (wt. %) and ρ_{water} is the density of water at 5 °C. Sediment porosity (φ) was calculated as:

$$\varphi = \frac{\rho_{wet}}{\rho_{water}} \times w$$

and tortuosity (θ) was calculated as:

$$\theta = \sqrt{1 - \ln \varphi^2}.$$

The diffusion coefficients (D_i) for different temperatures at the SWI were calculated using the Stokes–Einstein relation:

$$D_i = D_{i,25^\circ C} \times \frac{\mu_{25^\circ C} \times T}{\mu_T \times 298.15 K},$$

where $D_{i,25^\circ C}$ is the diffusion coefficient of ion i at 25 °C, $\mu_{25^\circ C}$ and μ_T are the dynamic viscosities of water at 25 °C (0.890 g m⁻¹ s⁻¹) and ambient temperature, T , respectively.

Given a pH of Lake Stechlin typically slightly above 8, equimolar concentrations of H₂PO₄⁻ and HPO₄²⁻ were assumed. Diffusion coefficients for H₂PO₄⁻ (8.46 cm² s⁻¹) and HPO₄²⁻ (7.34 cm² s⁻¹) at 25 °C were taken from Li & Gregory (1974). SRP fluxes were calculated separately for each basin and for both June and October 2021. The annual SRP flux from the entire basin sediment surface was calculated by multiplying the mean P fluxes determined for each depth layer in each of the three lake basins by the corresponding sediment surface area.

The SRP fluxes from the sediment to the water column were also compared with hypolimnetic TP accumulation (HPA) rates calculated from the increase in TP concentrations for the early (1 May–31 July) and late (1 August–31 October) summer stratification period in 2021. The increase in TP concentration in a given water layer was multiplied by the volume of the depth interval and divided by the corresponding sediment surface area and the elapsed time.

Differences in SRP contents and fluxes among lake basins, between months and with water depth were

analyzed by using general linear models with lake basin and month as categorical variables and depth as a continuous variable. All interactions were included in the models. The analysis of TP content followed the same approach but did not include month, because TP data were available only for June. We also used a model with lake basin and oxygen depletion as factors and depth as a covariate to test whether oxygen depletion was related to P fluxes. Finally, we tested for differences in P fluxes among the three lake basins from the deepest sediments where replicate samples were available, as well as among the shallowest sediments in the littoral zone (5 and 15 m) in each of the three lake basins. All statistical analyses were performed using SYSTAT version 13.1.

Results

Soluble reactive phosphorus concentrations

Porewater SRP concentrations in surface sediments of Lake Stechlin significantly increased with water depth (Fig. 1b–d; $p < 0.0001$). Slopes differed neither among lake basins ($p = 0.75$) nor between months ($p = 0.15$), nor were any of the interactions significant ($p = 0.21$ to 1.00). Nevertheless, in the deepest areas of all three lake basins, porewater SRP concentrations were always higher in October than in June (Fig. 1b–d), whereas the opposite was found for most of the shallower sites (5–25 m depth). In the main basin, the highest porewater SRP concentrations were 1.36 mg L⁻¹ at 35 m depth in June and 1.65 mg L⁻¹ at 65 m in October (Fig. 1b). In the west basin, peak concentrations were 0.82 mg L⁻¹ at 25 m in June and 1.58 mg L⁻¹ at 40 m in October (Fig. 1c). The highest values in the south basin (25 m) were 0.66 mg L⁻¹ in June and 1.19 mg L⁻¹ in October (Fig. 1d). Despite the strong influence of depth, which accounted for 80% of the explained variation in porewater SRP concentration, more than 70% of the total variation remained unexplained. For example, concentrations at the four 65 m sites sampled in the main basin ranged from 0.64 to 1.65 mg L⁻¹ in October (Fig. 1b). Likewise, porewater SRP in sediments at 25–35 m depth sometimes showed concentrations as high as in the deepest parts of the lake (e.g., 1.36 mg L⁻¹ at

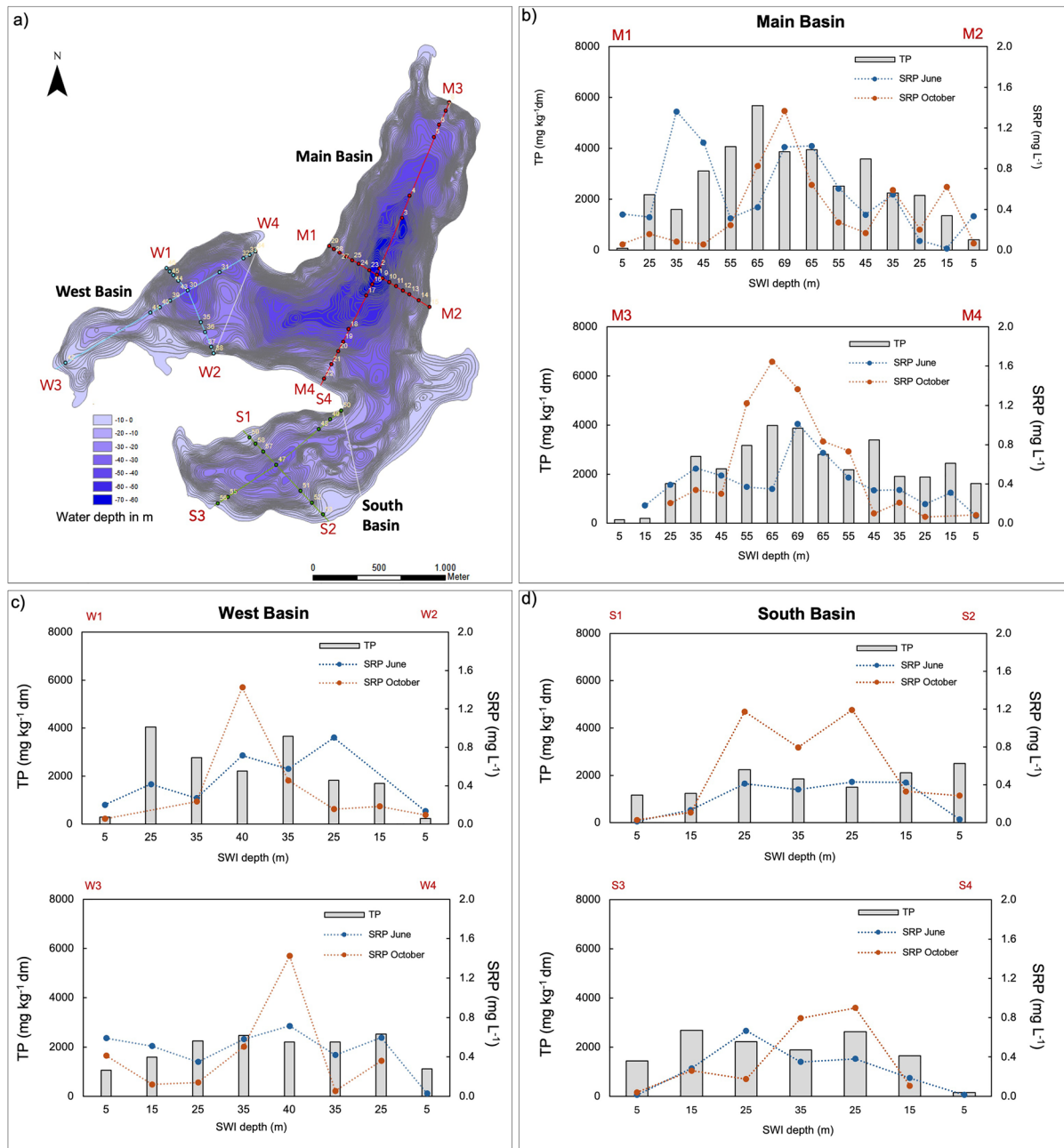


Fig. 1 a Location of 54 sites in Lake Stechlin where sediment cores were taken; **b–d** sediment total phosphorus (TP) concentrations in June (gray bars) and pore water SRP concentrations

at a sediment depth of 2 cm along transects in the **b** main, **c** west, and **d** south basin of Lake Stechlin in June (blue line) and October (orange line). ND = no data

35 m vs. 1.02 mg L⁻¹ at 69 m in the main basin in June; Fig. 1b).

Sediment total phosphorus concentrations

Concentrations of TP in surface sediments of the main basin of Lake Stechlin ranged from 75 mg kg⁻¹ dry mass (dm) at 5 m depth to 5370 mg kg⁻¹ at 65

m (Fig. 1b). In the west basin, TP concentrations ranged between 230 mg kg^{-1} at 5 m and 4000 mg kg^{-1} at 25 m depth (Fig. 1c). In the south basin, TP concentrations ranged from 157 mg kg^{-1} at 5 m to 2690 mg kg^{-1} at 15 m (Fig. 1d). Like for porewater SRP, sediment TP concentration strongly increased with water depth ($p < 0.001$, $N = 54$), accounting for as much as 97% of the total explained variation, whereas difference among lake basins were not significant ($p = 0.35$), nor was there a significant interaction between depth and lake basin ($p = 0.62$). Most of the total variation (62%) remained unexplained (Fig. S1, 1b–d).

Soluble reactive phosphorus fluxes

The spatio-temporal pattern of SRP fluxes across the SWI closely resembled the pattern of porewater concentrations (Fig. 2a–d). Fluxes increased with water depth ($p < 0.001$, $N = 105$), which accounted for 75% of the total variation, but were similar in all three basins ($p = 0.58$) and also between June and October

($p = 0.13$), with none of the interactions between water depth, lake basin and sampling month being significant, either ($p = 0.25$ to 1.00). As with the concentration of porewater SRP, SRP fluxes showed high variation independent of the depth gradient, resulting in 76% of the total variation being unexplained. However, the average SRP flux across the SWI in October was higher at depths where oxygen was completely depleted ($p < 0.001$), a pattern that was consistent across the three lake basins (Figs. 2b–d, 3b, 4). In contrast, no difference in fluxes was observed at the same water depth in June ($p = 0.98$), when the hypolimnion in the main basin was well oxygenated from the surface to the lake bottom (Fig. 3b).

Vertical profiles of physico-chemical conditions in the water column (main basin)

In June, the epilimnion extended to a depth of 4 m at a water temperature of 24°C , which sharply declined in the metalimnion to drop below 5°C at 15 m (Fig. 3). By October, the epilimnion had deepened to 9 m at a

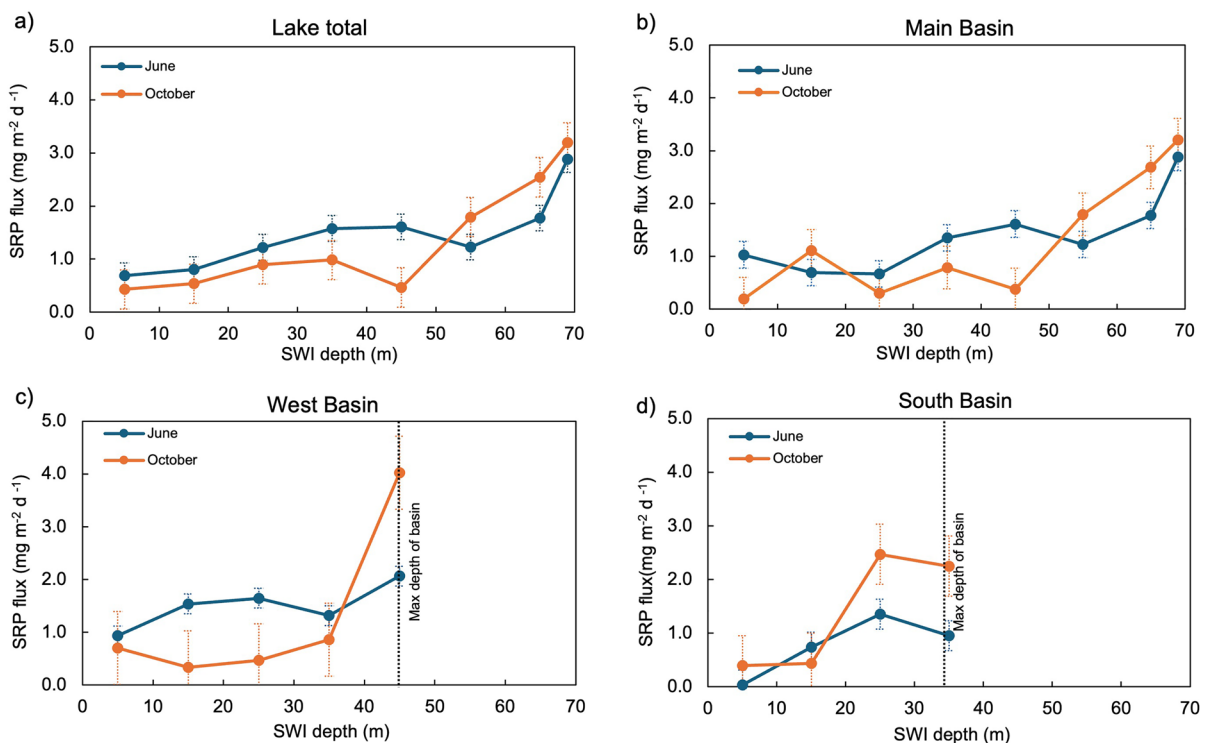


Fig. 2 Mean SRP fluxes at different water depths in the **a** lake as a whole, **b** main, **c** west and **d** south basin in June (blue line) and October (orange line) 2021; error bars represent standard deviations

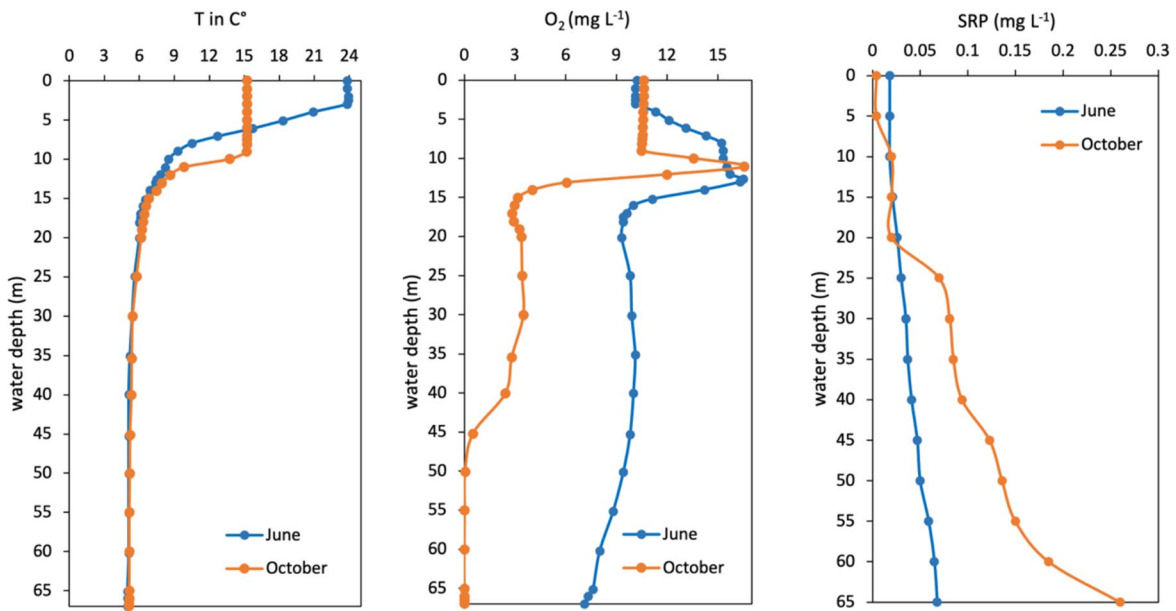


Fig. 3 Vertical profiles of temperature, oxygen concentration and SRP concentration in the water column of the main basin of Lake Stechlin on 22 June (blue lines) and 7 October (orange lines) 2021

temperature of 15 °C. In June, oxygen concentrations of 10.2 mg L⁻¹ (120% saturation) in the epilimnion increased to a metalimnetic maximum of 16.5 mg L⁻¹ (138%) at 13 m depth, followed by a metalimnetic minimum of 9.4 (76%) at 18 m to about 7.1 mg L⁻¹ at 67 m and little change in the hypolimnion (Fig. 3). This oxygen profile contrasts with that in October when the metalimnetic maximum and minimum occurred at lower depths and, in particular, oxygen depletion characterized the hypolimnion, resulting in anoxic conditions below 50 m depth. Finally, SRP concentrations in June gradually increased from 0.018 mg L⁻¹ in the epilimnion to 0.068 mg L⁻¹ at 65 m depth. By October, these concentrations had decreased at the water surface (0.004 mg L⁻¹) but increased in the hypolimnion to reach 0.260 mg L⁻¹ at 65 m (Fig. 3). In June, surface pH was 8.7, rising to 9.0 at 13 m before dropping to 7.6 at 15 m and 7.5 at 65 m. In October, pH was 8.6 at the surface, peaked at 9.2 at 11 m, then declined to 7.1 below 15 m, remaining near 7.0 down to 65 m.

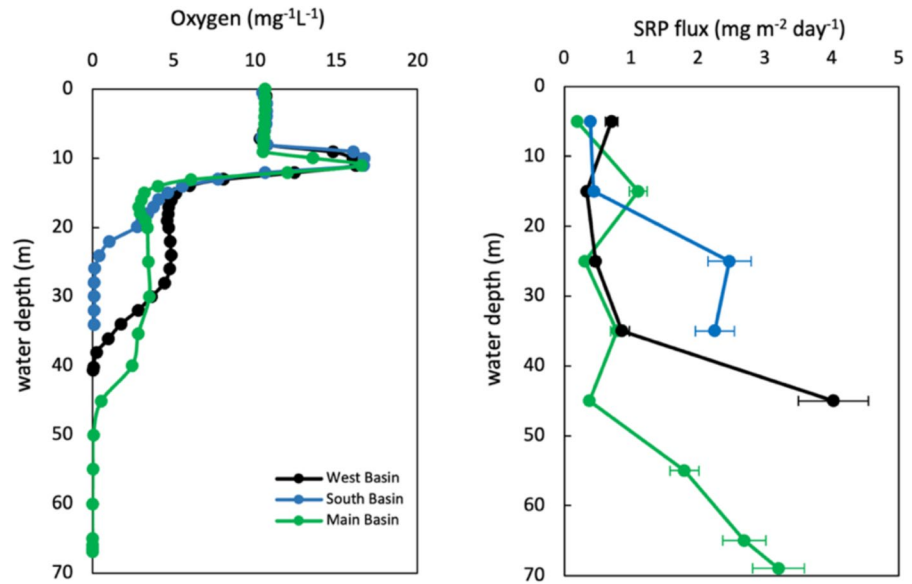
Similar depth gradient trends were observed in the other sub-basins. However, routine depth gradient measurements were only recorded in the main basin during the same period as the coring campaign in June and October. Profiles for the south and west

basins are available for July and September 2021 (see SI Figs. S3 and S4).

Discussion

Porewater SRP is the most mobile fraction in the sediment P pool and hence most sensitive to changing environmental conditions (Boström et al. 1988; Nowlin et al. 2005; Syers et al. 1973). Consequently, knowledge on porewater SRP concentrations is crucial to understand P dynamics in and release from sediments (Enell and Löfgren 1988). Attempts to predict P release rates from lake sediments based on correlations with sediment TP concentrations across lakes (Nürnberg 1988) were unsuccessful for Lake Stechlin where SRP fluxes proved not to be positively correlated with sediment TP concentrations (Fig. S3). Clearly, knowledge on sediment TP concentrations is not always sufficient to infer P fluxes across the SWI. This conclusion may particularly apply to lakes where sediment properties are influenced by a complex interplay between sediment deposition, resuspension, and transport, resulting in considerable spatial heterogeneity (Mackay et al. 2012). The TP concentration of sediments generally increases with

Fig. 4 Depth profiles of O_2 concentrations and SRP fluxes across the sediment water interface in the main (green), west (black) and south (blue) basin of Lake Stechlin in October 2021



water depth, due to the focusing of fine-grained sediments in deeper parts of lake basins (Albright et al. 2022; Cardoso-Silva et al. 2018; Mackay et al. 2012). However, P release from sediments is not only contingent on the sediment TP concentration, but also depends to a large extent on the P binding forms and the biogeochemical conditions in the sediment (Boström et al. 1988; Enell and Löfgren 1988; Orihel et al. 2017). Therefore flux determinations in different parts of a lake are required to address the P release and its spatio-temporal variability (Lewandowski et al. 2002).

At depths greater than 45 m, elevated mean SRP fluxes in October (Fig. 2a) could be the result of hypolimnetic oxygen depletion at the end of summer stratification. Anoxia can induce the release of P bound to Fe and Mn (Einsele 1936; Hupfer and Lewandowski 2008; Mortimer 1941). Indeed, this P fraction accounts for a substantial portion of sediment P in Lake Stechlin (Gonsiorczyk et al. 2003), suggesting potential for redox-sensitive P release when oxygen is depleted. Elevated P release rates associated with near complete oxygen depletion at much shallower water depths in the south (> 25 m) and west basins (> 40 m) than in the main basin (> 50 m) indicate that redox-controlled P release could indeed be an important mechanism (Fig. 4). This conclusion is consistent with the increase of hypolimnetic SRP concentrations towards the lake bottom that we observed in all three lake basins

in October. However, P accumulation and oxygen depletion in deep water often occur concomitantly and could both result also from the mineralization of settling organic matter.

Remarkably, the total P flux across the SWI for the entire lake was 32% higher in June (1736 kg yr^{-1}) than in October (1182 kg yr^{-1}) when anoxic conditions had greatly expanded in the hypolimnion. This seemingly paradoxical finding is mainly due to the large extent of relatively shallow lake areas (Table 1). Specifically, areas with water depths < 20 m accounted for 46% of the total sediment surface, whereas areas with a water depth of more than 50 m accounted for only 8%. Thus, although redox-mediated P-release from sediments near the bottom of the three lake basins may play an important role, our data also highlight another, previously underrated SRP flux from sediments in relatively shallow lake areas (Table 1, Fig. 2a). This additional P source, driven by enhanced remineralization of organic P under aerobic conditions, contributes notably to internal loading.

The importance of the littoral zone for P fluxes from sediments to the water column, and hence lake eutrophication, has been previously proposed for Lake Stechlin (Gonsiorczyk et al. 2024). The exact causes are not entirely clear, but long-term monitoring data and mass balance models suggest that the depletion of iron (Fe) as a binding agent for sedimentary P may be a contributing factor (Gonsiorczyk et al. 2024), and a

major shift in the submerged macrophyte community from characeae to annual vascular plants also appears to have an influence (Hilt et al. 2025). In nutrient-poor hardwater lakes, charophytes dominate submerged vegetation and trap P in the lakebed, maintaining low surface P and high-water clarity. When replaced by vascular macrophytes, which release P after summer, P levels are likely to rise during winter, boosting phytoplankton in next spring and reducing water clarity, favouring in turn higher vascular plants over charophytes (Hilt et al. 2025).

Furthermore, the release of co-precipitated P due to calcium carbonate dissolution can play an important role in P cycling in freshwater ecosystems (Corman 2025) and may have contributed to the sedimentary P fluxes in Lake Stechlin as well. Carbonate dissolution in sediments of Lake Stechlin might be facilitated by changes in sediment pore water chemistry due to intensified remineralization.

Unlike the seasonally anoxic deep zones of the lake, littoral sediments are continuously exposed to an oxygenated water column, which tends to enhance P release as a result of organic matter remineralization at the SWI (e.g., Boström et al. 1988). Thus, although the exact mechanisms driving P release from sediments in shallow lake areas are not fully understood, our results point to an important role of these areas in the lake's phosphorus cycle.

Water-column processes could also contribute to the recent eutrophication trend in Lake Stechlin along with sedimentary P release. Increased carbonate deposition in surface sediments (Scholtysik et al. 2020) coincides with increased primary production (Kröger et al. 2023) over the past two decades. With

Table 1 P fluxes across the sediment–water interface between 0–20 m, 20–50 m, 50–69 m and 0–69 m water depth in June and October 2021

Month	Depth layer (m)	Area (m ²)	Mean P flux (kg yr ⁻¹)	Mean P flux (g m ² yr ⁻¹)
June	0–20	1,959,574	539	0.28
	20–50	1,915,936	1011	0.53
	50–69	356,039	187	0.53
	0–69	4,231,549	1737	0.41
October	0–20	1,959,574	347	0.18
	20–50	1,915,936	576	0.30
	50–69	356,039	260	0.73
	0–69	4,231,549	1183	0.28

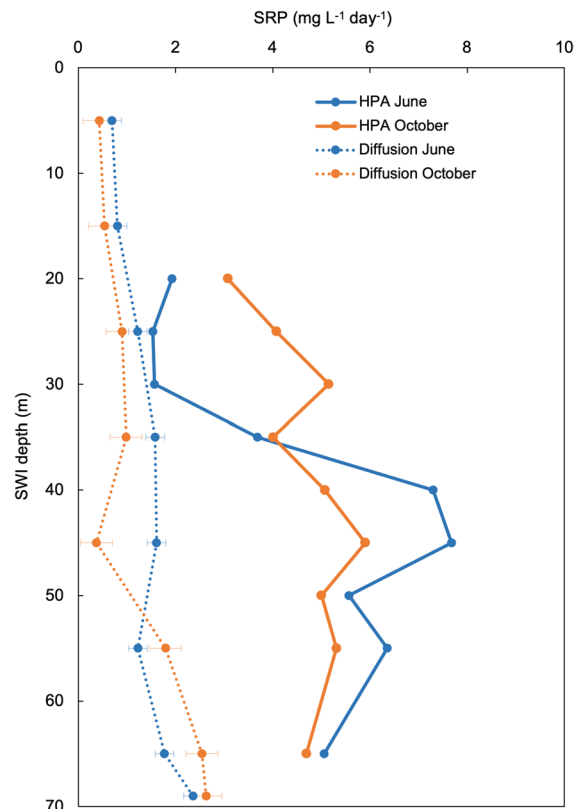


Fig. 5 Comparison of hypolimnetic P accumulation rates (HPA, dotted lines) with calculated SRP fluxes (solid lines) in June (blue lines) and October (orange lines) of 2021

a maximum depth of nearly 70 m, mineralization of sinking organic matter in both oxic and anoxic conditions conceivably drove substantial SRP release in the water column. In support of this hypothesis, hypolimnetic P accumulation rates far exceeded SRP release rates (Fig. 5) suggesting a 'short-circuited' P cycling. Rapid oxygen depletion leading to anoxic conditions below 45 m by October (Fig. 3b) suggest indeed that biomass remineralization in the water column of the main lake basin was intense. Given that P accumulation rates in the water column increased with water depth, especially in October, with peaks at 45 or 50 m depth in the main basin, these depth gradients appear to be due to two complementary processes: (i) the mineralization of settling organic particles within the water column and (ii) diffusive SRP fluxes from the sediments.

Conclusions

Sediment TP concentrations had no predictive power to infer spatial and temporal patterns of SRP fluxes observed across, and within, the three lake basins of Lake Stechlin. Instead, hypolimnetic redox processes in or near deep-water sediments appear to be an important mechanism driving P fluxes across the SWI. Nevertheless, high SRP fluxes from sediments may also occur in oxic conditions at much lower depths. This suggests an important role of shallow lake areas in releasing P to the water column and that hypolimnetic oxygen depletion leading to anoxic conditions is not imperative for major internal P release to occur. In fact, given the large sediment surface of shallow lake areas (0–20 m), its summed contribution to the total areal SRP flux across the SWI was much larger (up to threefold) than the total release from deep-water sediments (> 50 m). Clearly, shallow lake areas need to be adequately considered in sampling strategies aimed to assess the role of sediments as a source of P for the water column of deep clearwater lakes. Additionally, water-column data indicate that the mineralization of settling and settled organic matter contributes to P accumulation in the hypolimnion. This 'short-circuited' P cycling could have been fueled by increased primary production of Lake Stechlin in recent years.

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Author contribution The study was initiated by JH and conceptualized by MOG. The sampling programme was designed by all authors, and field work was carried out by JH and TG. The data analysis was performed by JH, TG, AR and MOG. All authors were involved in interpreting the data. JH and VT drafted the manuscript. All authors were involved in editing and revising the manuscript.

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Data availability The datasets generated and analysed during the current study are available in the GRO.data base: <https://doi.org/10.25625/V6JSJ1>.

Declarations

Conflict of interest The authors declare no competing interests.

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